NoSQL & Vector Database Creation Specification

Course: DBMS Mini Project (UE24CS395) • **Institution:** PES University (2026–2027)
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1. MongoDB Collection Schema Validation (collection-schema.json)

MongoDB collection investigation_documents enforces strict structural validation via $jsonSchema, mandating required fields while supporting polymorphic nested metadata and 64-dimensional dense semantic vectors.

```
{
  "$jsonSchema": {
    "bsonType": "object",
    "required": ["title", "document_type", "content", "created_at"],
    "properties": {
      "case_id": { "type": ["string", "null"], "description": "FK to cases.case_id" },
      "document_type": { "type": "string" },
      "title": { "type": "string", "minLength": 3, "maxLength": 250 },
      "content": { "type": "string" },
      "summary": { "type": ["string", "null"] },
      "related_entity_ids": {
        "type": "array",
        "items": {
          "type": "object",
          "required": ["entity_type", "entity_id"],
          "properties": { "entity_type": { "type": "string" }, "entity_id": { "type": "string" } }
        }
      },
      "tags": { "type": "array", "items": { "type": "string" } },
      "embedding": {
        "type": "array",
        "items": { "type": "number" },
        "description": "64-dimensional dense semantic vector for cosine similarity search"
      },
      "metadata": { "type": "object", "additionalProperties": true }
    }
  }
}
```

2. Indexes & Vector Embedding Architecture

- **Weighted Text Index:** doc_fulltext_index on { title: 10, tags: 5, summary: 3, content: 1 }.
- **Compound Index:** idx_case_id_created on { case_id: 1, created_at: -1 }.
- **Multikey Index:** idx_related_entities on { 'related_entity_ids.entity_type': 1, 'related_entity_ids.entity_id': 1 }.
- **64-Dim Dense Vector:** Pre-computed unit-normalized vectors stored in embedding for sub-millisecond cosine similarity search.